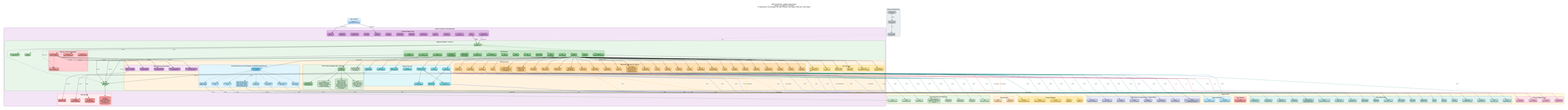

ABCT Architecture - A Better Crypto Tracker
Version 1.14.0 (Build 1771717923)
51 Blockchains | 42 Exchanges | 80+ DeFi Protocols | P&L Engine | Multi-User | Demo Mode

User Interface
Web Browser
(localhost:8000 / Port 8080)
HTTP/HTTPS GET /
Authentication
Docker Container: abct-dashboard
Frontend (Static Files)
index.html
Dashboard UI
wallets.html
Wallet Manager
transactions.html
Transaction History
nft-wall.html
NFT Gallery
login.html
Authentication
apis.html
API Manager
security.html
Security Settings
services.html
Health Status
backup.html
Backup & Restore
logs.html
System Logs
cache.html
Cache Manager
styles.css
Theming (5 themes)
app.js
Client Logic
transactions.js
Transaction Analytics
REST API
JSON
Transaction API
Backend (FastAPI + Uvicorn)
main.py
Application Entry
scans
nft_image_database.py
Image Cache DB
config.py
Configuration
session check
Security & Demo Middleware
demo_mode.py
Demo User Detection
rate_limit.py
Rate Limiting (slowapi)
size_limit.py
Request Size (10MB)
auth.py
Session Authentication
demo user
API Routers
settings
API Key Management
auth
Login/Logout/Sessions
security
SSL/Certificate Mgmt
exchanges
42 Exchange APIs
+ ExchangeRegistry
defiprotocols
Protocol Detection
+ P&L Analytics
portfolio
Summary & History
transactions
Multi-Chain Txn History
defi
DeFi Positions
prices
Crypto Prices
nfts
NFT Collections
custom-tokens
Token Tracking
pnl
Cost Basis
P&L Reporting
logs
System Logging
wallets
Wallet CRUD
backup
Export/Import
nft-scheduler
Background NFT Updates
ExchangeRegistry
BSC/ARB/AVAX NFTs
Demo Services (Fake Data)
demo_nft_service.py
76 NFTs, 7 Collections
demo_defi_service.py
Staking Positions
demo_wallet_service.py
91 Tokens, 26 Chains
demo_exchange_service.py
Exchange Balances
demo_transaction_service.py
1,500 Transactions
API keys
sessions
SSL config
log entries
SVG files
cert files
images
database.py
SQLite Layer
Exchange Services (42 Exchanges via BaseExchangeService)
base_exchange.py
BaseExchangeService
+ 5 Auth Mixins
35 more
bitget_service.py
Bitget Exchange
gate_service.py
Gate.io
kucoin_service.py
KuCoin
bybit, mexc, kraken, gemini
bitfinex, htx, bingx, phemex
woox, poloniex, cryptocom
bitstamp, bitmart + 22 more
(35 new exchange services)
coinbase.py
Coinbase CDP
binance_service.py
Binance.com
binance_us_service.py
Binance.US
okx_service.py
OKX Exchange
DeFi Protocol Adapters (80+ Protocols)
registry.py
ProtocolRegistry
(auto-discovery)
cost_basis_engine.py
FIFO/LIFO/Average
P&L Analytics
base_adapter.py
ProtocolAdapter ABC
5 DetectionMethods
cardano/
Minswap, SundaeSwap V3
WingRiders, Splash, Djed
FluidTokens, Lenfi, MuesliSwap
Liqwid, Indigo, Strike
Surf, Iagon (13 adapters)
evm/
Aave v3, Compound v3
Uniswap v3 LP, Curve
Balancer, EigenLayer
Maker, Morpho, GMX
Pendle, Stargate, Aerodrome
Velodrome, Radiant, Benqi
SushiSwap, Yearn v3, Beefy
+ liquid staking tokens
55+ protocols
solana/
Marinade, Jito, Orca
Raydium, Drift, MarginFi
Kamino, Jupiter Perps
BlazeStake + 6 more
15 adapters
Support Services
ssl_service.py
Certificate Mgmt
snapshot.py
Portfolio Snapshots
logging_service.py
Centralized Logs
taptools.py
Cardano Token Data
transaction_history.py
Multi-Chain Txns
defi.py
DeFi Protocols
pricing.py
Price Aggregation
moralis.py
Multi-Chain Fallback
logostream.py
Token Logo API
etherscan.py
ETH Transaction History
Services Layer
Blockchain Services (51 Chains)
vechain.py
VeChain (VET)
cosmos.py
Cosmos (ATOM)
near.py
NEAR (NEAR)
icp.py
ICP (ICP)
cosmos_chain.py
Cosmos IBC (config-driven)
Osmosis/Celestia/Injective
dYdX/Sei/Akash
substrate_service.py
Substrate Chains
Polkadot (DOT)
Kusama (KSM)
ton_service.py
TON (TON)
stellar_service.py
Stellar (XLM)
kaspa_service.py
Kaspa (KAS)
ergo/iota/waves
mina/zilliqa/kaia
(public REST APIs)
cardano.py
Cardano (ADA)
ethereum.py
Ethereum (ETH)
polygon.py
Polygon (MATIC)
base.py
Base (ETH)
evm_chain.py
Generic EVM Service
(BSC/Arbitrum/Avalanche
Optimism/zkSync/Linea
Scroll/Fantom/Cronos
Gnosis/Moonbeam/Kaia)
algorand.py
Algorand (ALGO)
tron.py
Tron (TRX)
bitcoin.py
Bitcoin (BTC)
solana.py
Solana (SOL)
xrp.py
XRP (XRP)
hedera.py
Hedera (HBAR)
multiversx.py
MultiversX (EGLD)
sui.py
Sui (SUI)
aptos.py
Aptos (APT)
filecoin.py
Filecoin (FIL)
litecoin.py
Litecoin (LTC)
dogecoin.py
Dogecoin (DOGE)
zcash.py
Zcash (ZEC)
tezos.py
Tezos (XTZ)
stacks.py
Stacks (STX)
NFT Services
nft_image_service.py
Image Caching
nft.py
Cardano NFTs
ethereum_nft.py
Ethereum NFTs
solana_nft.py
Solana NFTs
algorand_nft.py
Algorand NFTs (ARC-3/69)
nft_scheduler.py
Background Updates
Data Storage
static/demo-nfts/
Demo NFT SVGs
data/certs/
SSL Certificates
(server.crt, server.key)
nft_images.db
NFT Image Cache
(thumbnails, metadata)
portfolio.db
Main Database
(users, sessions, wallets,
balances, API keys, cache,
transactions, nfts,
security settings, logs)
HTTPS
HTTPS (35)
Cardano UTXOs
DeFi contract calls
Checkmol Staking
HTTPS (fallback)
HTTPS (BSC/ARB/AVAX)
Solana DeFi RPCs
1st
2nd
3rd
4th
Security & Monitoring
.git/hooks/pre-push
Git Integration
triggers
security_agent.py
Pre-push Hook
runs
security_audit.py
Automated Audit
External APIs
Exchange APIs (42 Platforms)
Bitget
Derivatives Exchange
Gate.io
Comprehensive API
KuCoin
Global Exchange
35 New Exchanges
Bybit, MEXC, Kraken, Gemini
Bitfinex, HTX, Crypto.com
+ 28 more
Coinbase CDP
Portfolio Sync
Binance.com
Global Exchange
Binance.US
US Exchange
OKX
Multi-Asset Exchange
Visual Assets
LogoKit
Blockchain Logos
Logostream
Token Logo API
Cardano Network
Blockfrost
Blockchain Data (Primary)
Maestro
Blockchain Data (Fallback)
TapTools
NFT Prices & Token Data
CExplorer
Staking Data
Koios
Fallback API
EVM Chains (12 via Alchemy + Public RPCs)
Snowscan
Avalanche Transactions
Etherscan
Ethereum Transactions
BscScan
BSC Transactions
Arbiscan
Arbitrum Transactions
Basescan
Base Transactions
Polygonscan
Polygon Transactions
Moralis
Multi-Chain Fallback
Alchemy
ETH/Polygon/Base/BSC/ARB/AVAX
+OPT/zkSync/Linea/Scroll
Algorand Network
Tatum
Algorand Data (Fallback)
Pera Wallet
Algorand Data (Primary)
Tron Network
TronGrid
TRX/TRC-20 Data (Free)
Other Blockchains
VeBlocks/Thor
VeChain Data (Free)
Cosmos LCD
PublicNode (Free)
NEAR RPC
+ NearBlocks (Free)
Rosetta API
ICP Data (Free)
Cosmos IBC LCD
Osmosis/Celestia/Injective
dYdX/Sei/Akash (Free)
Subscan
Polkadot/Kusama (opt. key)
TON Center
TON Data (opt. key)
Horizon API
Stellar Data (Free)
Mempool.space
Bitcoin Fallback
Blockstream
Bitcoin Data
Helius
Solana RPC & NFTs
XRPL Cluster
XRP Ledger API
Mirror Node
Hedera API
MultiversX API
MultiversX Data
Sui RPC
Sui Data
Aptos API
Aptos Data
Glif RPC
Filecoin Data
BlockCypher
LTC/DOGE Data (Free)
Blockchair
Zcash Data (Free)
TzKT
Tezos Data (Free)
Hiro API
Stacks Data (Free)
Pricing (Fallback Chain)
4. DefiLlama
Universal Fallback
1. CoinGecko
Primary Price Data
3. Coinbase
Spot Prices (Free)
2. CoinMarketCap
Secondary Prices